

Disk Drive Performance Activity — Answers

Based on Module 1: Introduction to Information Storage & Module 2: Data Center Environment

Short Answer

1. In your own words, define seek time.

Seek time (also called access time) is the time it takes for the disk's read/write (R/W) head to move radially across the platter and position itself over the correct track or cylinder where data needs to be read or written. The lower the seek time, the faster the I/O operation.

2. Why does random I/O suffer more from seek time than sequential I/O?

In sequential I/O, requested data is located in adjacent tracks/sectors, so the R/W head barely has to move between consecutive requests, keeping seek time minimal. In random I/O, each request may point to a completely different, non-adjacent cylinder, forcing the actuator arm to travel a significant and unpredictable distance for every single I/O. Because seek time is a mechanical, time-consuming operation, this repeated long-distance movement adds up and dominates the service time for random I/O, dragging down overall performance.

3. What is short-stroking, and why would an administrator use it?

Short-stroking is the practice of restricting a disk drive to use only a subset of its available cylinders (typically the outer ones) instead of the full disk — for example, using only the first 30–40% of cylinders on a drive. Because the R/W head then only has to travel across a smaller portion of the platter, the average seek distance (and therefore average seek time) is reduced. Administrators use this technique to increase IOPS and lower response time for performance-sensitive applications, accepting reduced usable capacity as the trade-off.

Problem Solving (Capacity Loss from Short-Stroking)

1. A 1 TB (1000 GB) disk drive is short-stroked to use only the first 30% of its cylinders.

- a) Usable capacity = $1000 \text{ GB} \times 0.30 = 300 \text{ GB}$
- b) Wasted (unusable) capacity = $1000 \text{ GB} - 300 \text{ GB} = 700 \text{ GB}$

2. Service Time and IOPS Calculation

Given: Full-stroke seek time = 20 ms | Average seek time (full drive) = 8 ms | Average seek time (short-stroked to 40%) = 3 ms | Rotational latency = 2 ms | Data transfer time = 0.5 ms

Formula:

Disk Service Time = Seek Time + Rotational Latency + Data Transfer Time

IOPS = $1000 / \text{Service Time (ms)}$

Configuration	Service Time (ms)	IOPS
Full Drive	$8 + 2 + 0.5 = 10.5 \text{ ms}$	$1000 / 10.5 \approx 95.2 \text{ IOPS}$
Short-Stroke (40%)	$3 + 2 + 0.5 = 5.5 \text{ ms}$	$1000 / 5.5 \approx 181.8 \text{ IOPS}$

Note: The 20 ms full-stroke figure is the worst-case (edge-to-edge) seek spec and is not used in this average-service-time calculation — only the given average seek times apply.

Compare

An administrator has two identical 600 GB drives: Drive A (full drive, all cylinders used) vs. Drive B (short-stroked to use only 50% of cylinders).

Question	Drive A	Drive B
Usable capacity	600 GB	300 GB
Relative seek distance	Long	Short
Expected IOPS	Lower	Higher
Best suited for	Capacity-intensive, less performance-sensitive workloads (e.g., archiving, backups, large file storage)	Performance-sensitive, high-IOPS workloads (e.g., OLTP databases, transaction logs, latency-critical apps)

Why: Drive B sacrifices half its capacity to keep the R/W head confined to a narrower band of cylinders, minimizing seek distance and boosting IOPS/response time — the classic short-stroking trade-off of capacity for performance. Drive A maximizes usable space but pays for it with longer average seeks and lower IOPS.